

Daniel Lu

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Education

Carleton University, B.Sc. in Computer Science, Honours Program (Co-op) Ottawa, Canada
AI & Machine Learning Stream - Dean's Honours List - GPA: 3.9/4.0 *Sep 2023 – Present*

- **COMP 4107** (Advanced Neural Networks, Transformers, Model Optimization)
- **COMP 3105** (Supervised/Unsupervised Methods, Gradient Boosting, DNNs)
- **COMP 3106** (LLMs, RAG, Agentic Systems, Embeddings, NLP, Graph & Vector Databases)

Experience

Software Developer May 2026 – Aug 2026
Shopify *Toronto, Canada*

- Incoming Summer 2026

Software Developer May 2025 – Aug 2025
TrendAI *Ottawa, Canada*

- Led development of a **digital twin PoC**, creating a **RAG system** with **LangChain**, **Neo4j**, **PGVector**, and **Model Context Protocol (MCP)** for predictive infrastructure security analytics.
 - * Delivered an executive presentation to **TrendAI CEO Office** in Taiwan; pitched its technical value to an audience of **70+** and secured approval from **C-suite stakeholders**.
- Enhanced system performance and scalability for **120+ active enterprise users** by architecting cloud infrastructure optimizations across **AWS/Azure** with **Kubernetes HPA** and **pod load balancing**.
- Engineered **CI/CD pipelines** for internal SDKs and database deployment systems serving **200+ developers** and **10,000+ stakeholders**, reducing release cycles by **65%** from a whole day.

Software Developer Jan 2025 – Apr 2025
Blackberry QNX *Ottawa, Canada*

- Engineered **core automation libraries** in **Python** for the **QNX RTOS**, expanding code coverage by **15%** in a safety-certified environment.
- **Boosted shipping speeds** by **20%** by developing robust **internal tools** and scripts in **Python, C, and Bash** to optimize system validation.

Projects

Tobio | [Website](#) | [GitHub](#) Jan 2026

- Deployed a **monocular 3D reconstruction pipeline** that transforms standard video into **professional-grade spatial analytics**, democratizing elite tracking tools without the need for expensive sensor arrays.
- Slashed analysis time by **85%** by implementing **automated event recognition** and a **Gemeni RAG agent**, allowing coaches to uncover complex strategic patterns simply by asking questions.
- Achieved **sub-centimeter tracking precision** under **severe visual occlusion** by engineering a resilient **computer vision system** (YOLOv11 + ReID), enabling real-time 3D mapping of player movements and high-velocity ball trajectories.

CloudQ | [Devpost](#) | [GitHub](#) Jan 2025

- Won **Best Web 3.0 Hack** at **McHacks 12** (competing against **500+ participants**).
- Decreased queue times by more than **50%** by deploying a **decentralized, virtual queue management system** powered by **blockchain and RAG technology**, ensuring **efficient line-free queues** for events.
- Developed a modern, responsive UI using **React.js**, **Vite**, and **TailwindCSS**, integrating **Auth0**, **Starknet**, and **Hugging Face** for authentication, smart contracts, and AI-powered features.

Technical Skills

Languages: Python, C/C++, Ruby, Java, JavaScript, TypeScript, Haskell, Rust, SQL

Libraries and Frameworks: TensorFlow, PyTorch, Scikit-learn, LangChain, PGVector, Neo4j, Node, Express

Platforms & Technologies: AWS, Azure, Docker, Kubernetes, Hugging Face, Jenkins, Terraform